

Total No. of Questions : 6]

SEAT No. :

P63

[Total No. of Pages : 2

BE/Insem./APR-101

B.E. (Civil)

401007 : DAMS AND HYDRAULIC STRUCTURES

(2012 Pattern) (Semester - II)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q1 or 2, Q3 or 4, Q5 or 6.
- 2) Figures to the right indicate full marks.

Q1) a) Write short note on : **[5]**

- i) Dams & earthquakes
- ii) Displacement & Rehabilitation

b) Classify dams based on material use, purpose & Hydraulic action. **[5]**

OR

Q2) a) Write short note on : **[5]**

- i) Pendulum
- ii) Distributed fiber optic temp. tool.

b) Briefly explain any four instruments used for health monitoring of Dam. **[5]**

Q3) a) Explain in details any one forces causing instability of gravity dam. List any five causing instability of gravity dam. **[5]**

b) What is the limiting height of gravity Dam. **[5]**

OR

Q4) a) A 20 m high concrete gravity dam has vertical upstream face and down stream face is inclined at 45° . The top and base widths are 2 m and 20 m respectively. The free board is 2m. Take weight density of water as 10 kN/m^3 & of concrete 24 kN/m^3 Determine the factor of safety against overturning by neglecting earthquake and tail water. Consider full uplift. **[8]**

b) Define Arch Dam & Write two types of arch Dam. **[2]**

P.T.O.

- Q5)** a) State various types of spillway gates. At least five and its applications. [5]
- b) Draw a proportionate and neat sketches and explain function of each energy dissipator. [5]
- i) Roller Bucket
- ii) Ski jump bucket

OR

- Q6)** a) English Design steps of Ogee spillway from heel to toe and explain. [8]
- b) List four types of spillway. [2]

